



OCI Email Delivery Service

Deep Dive

Operations Advisory Team

Office of CTO | EMEA Technology Engineering

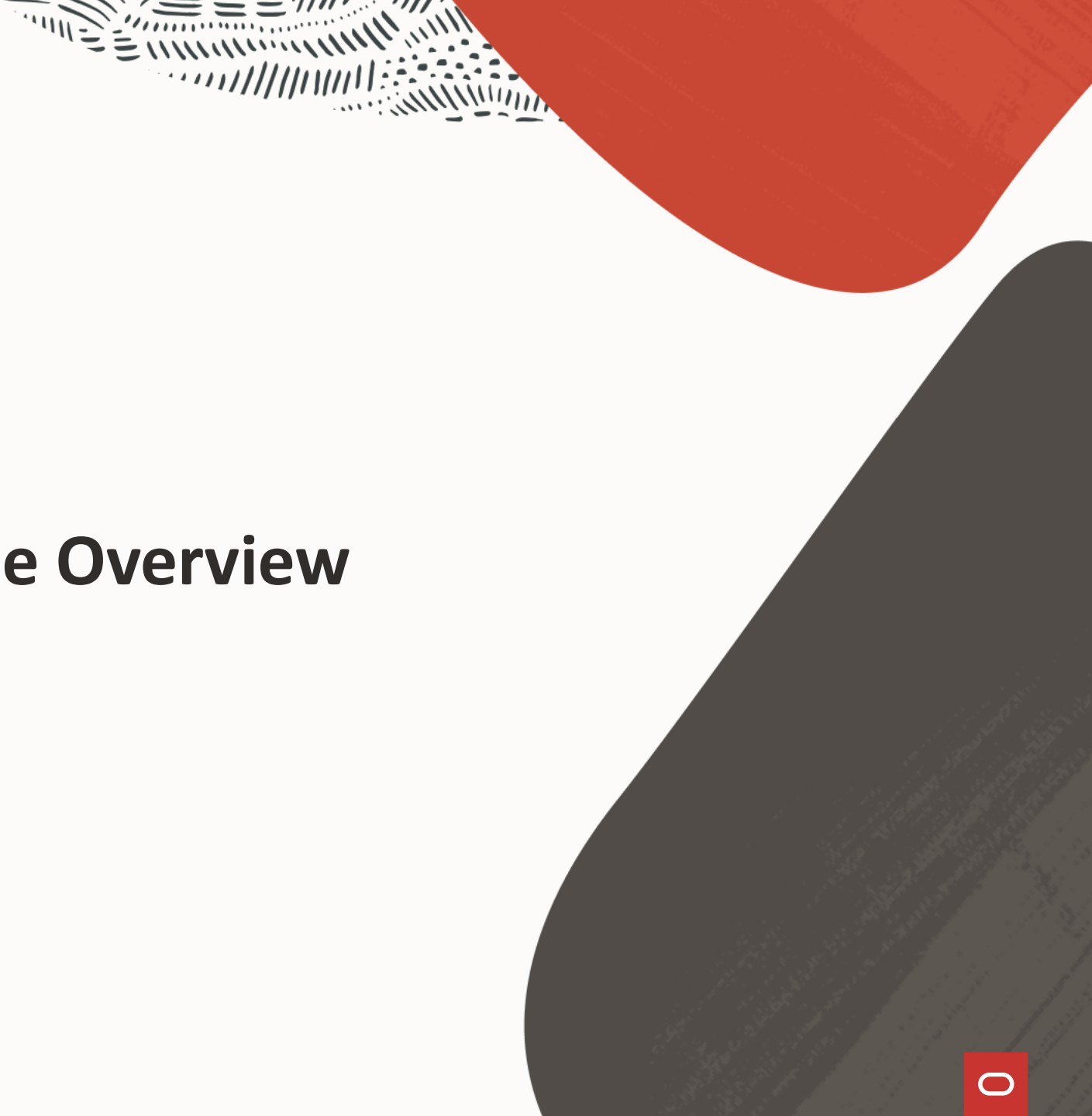
January 2026

Safe harbor statement

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Agenda

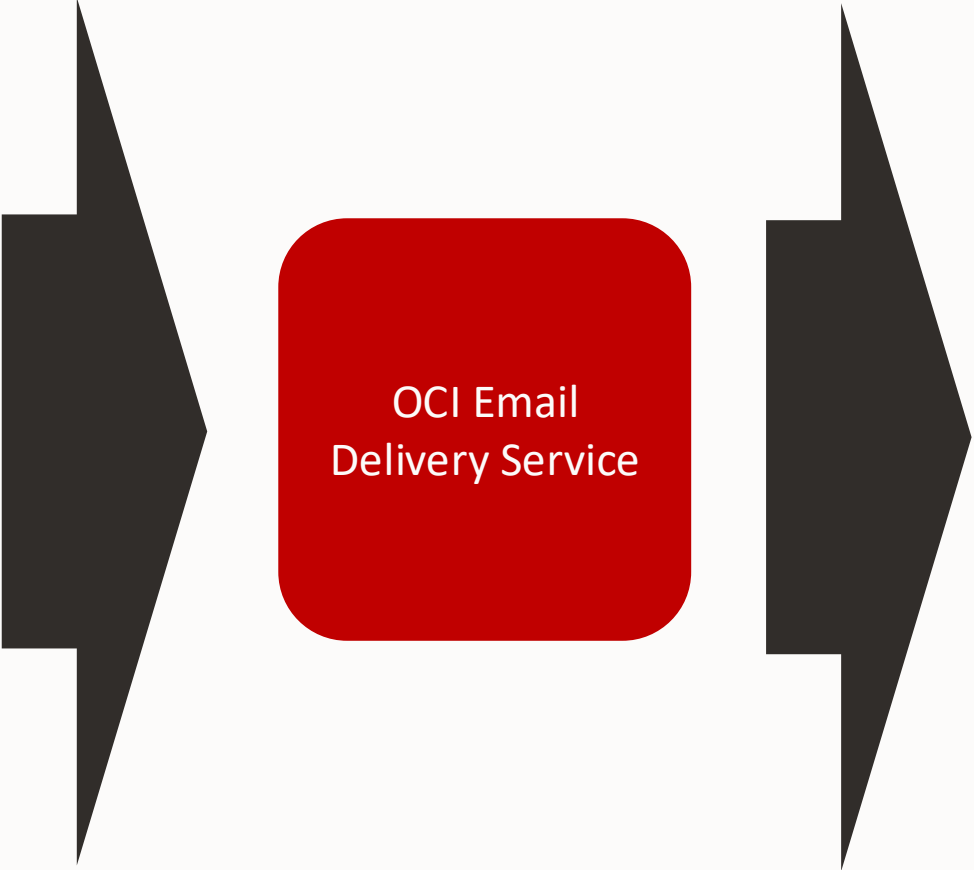
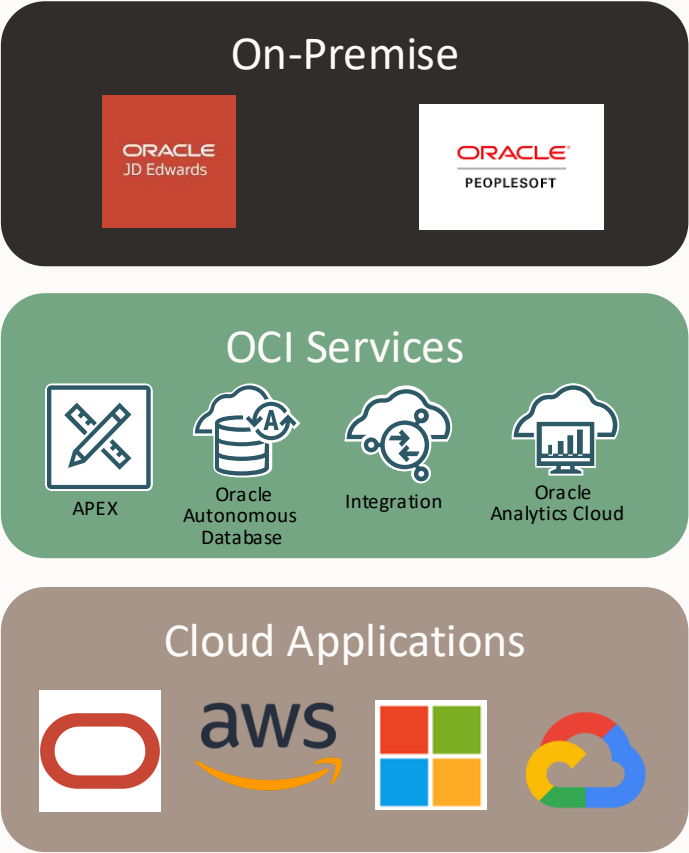
1. Overview of Email Delivery Service
2. IAM & DNS
3. Setting up email submissions
4. Managing email domains, approved senders, suppression list, DKIM and custom return path
5. Deliverability & Reputation — Governance Dashboard
6. Limits counting & cost



OCI Email Delivery Service Overview

OCI Email Delivery Service

*A single solution that reaches everyone...
everywhere*



- 
Customers
- 
Prospects
- 
Operations
- 
Business Stakeholders



What it is/What it isn't

✓	Managed outbound email sending at scale	A mailbox service (no IMAP/POP/receiving)	✗
✓	For transactional (OTP, password reset, receipts) and opt-in bulk mail	A full campaign manager (no list/segment/A/B UI built-in)	✗
✓	Integrates via HTTPS API (preferred) or SMTP with SMTP credentials	Required path for OCI Alarms/Events → use OCI Notifications → Email instead	✗
✓	Requires Approved Sender(s) or Verified Email Domain(s)	A relay for unsolicited/cold outreach	✗
✓	Auth & trust: DKIM (with Email Domains), plus SPF/DMARC recommended	A way to receive or host MX records	✗
✓	Regional quotas & default ~2 MB per message (raiseable); focus on deliverability	A place for large attachments → use links (e.g., Object Storage)	✗
✓	Regional sending for GDPR & CASL compliance		

Features



Managed Service



Secure Transport



Authorized Sending



Delivery Telemetry



Recipient Controls



Reputation Options



Regional Resources



Submission Paths



IAM & DNS

IAM Policies (Who can do what)

Roles

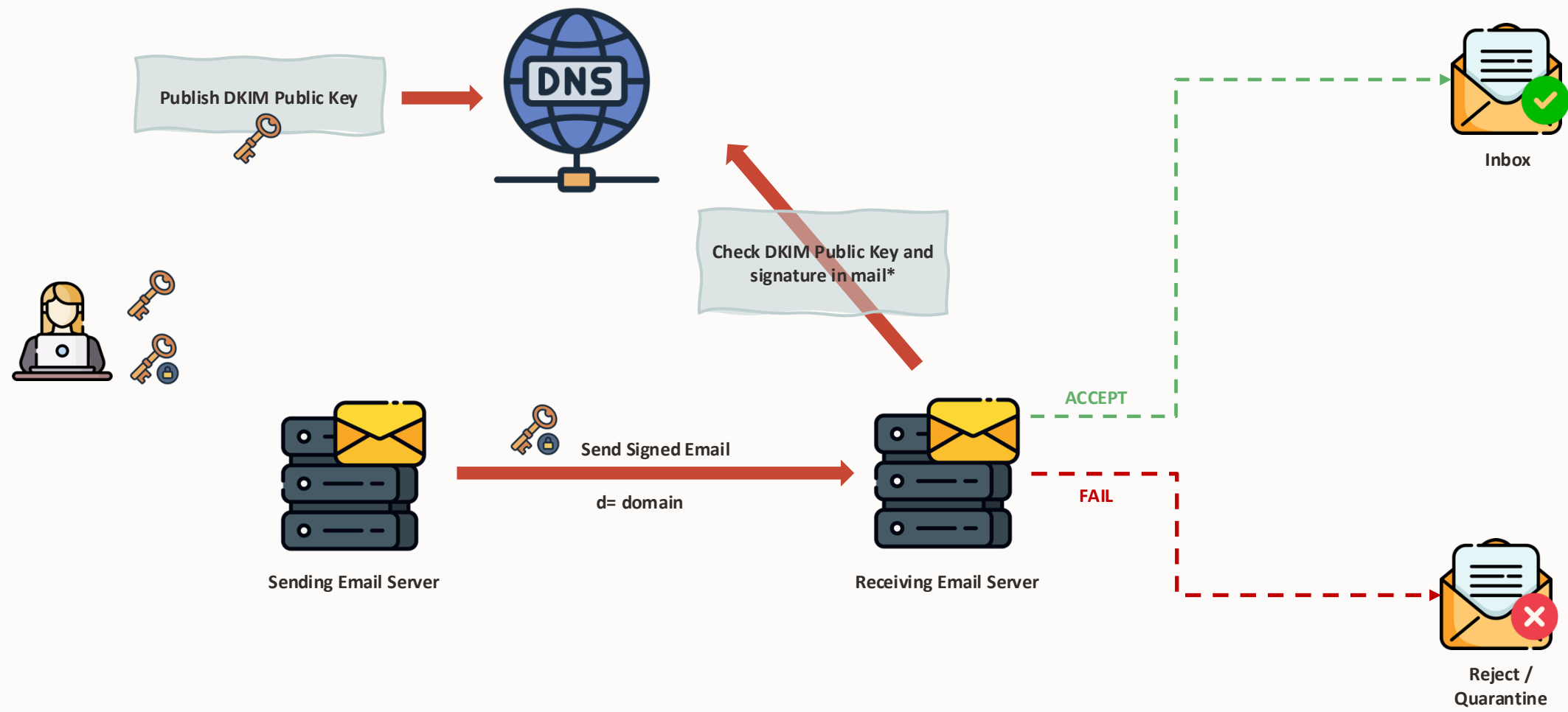
- **Admins (setup/governance)**
 - Grant "**manage email-family**" in the target compartment (approved senders, suppressions, email domains/DKIM).
 - Optionally restrict to "read/use" for least privilege.
- **To set up email submissions**
 - Grant "**manage email-family**" to the EmailDelivery service in the tenancy/compartment
- **App senders (HTTPS or SMTP)**
 - Put users in a group (e.g., 'email-senders') or use a dynamic group for instance principals.
 - Grant "**use email-family**" in the compartment where your Email Domain / Approved Senders live.

```
'Allow group email-senders to use email-family in compartment <COMPARTMENT_NAME>'
'Allow group email-admins to manage email-family in compartment <COMPARTMENT_NAME>'
'Allow dynamic-group app-servers to use email-family in compartment <COMPARTMENT_NAME>'
```

➤ SMTP credentials (if using SMTP)

- Create SMTP credentials on the IAM user (default 2 per user).
- Use host "smtp.email.<region>.oci.oraclecloud.com:587" with STARTTLS (TLS 1.2).

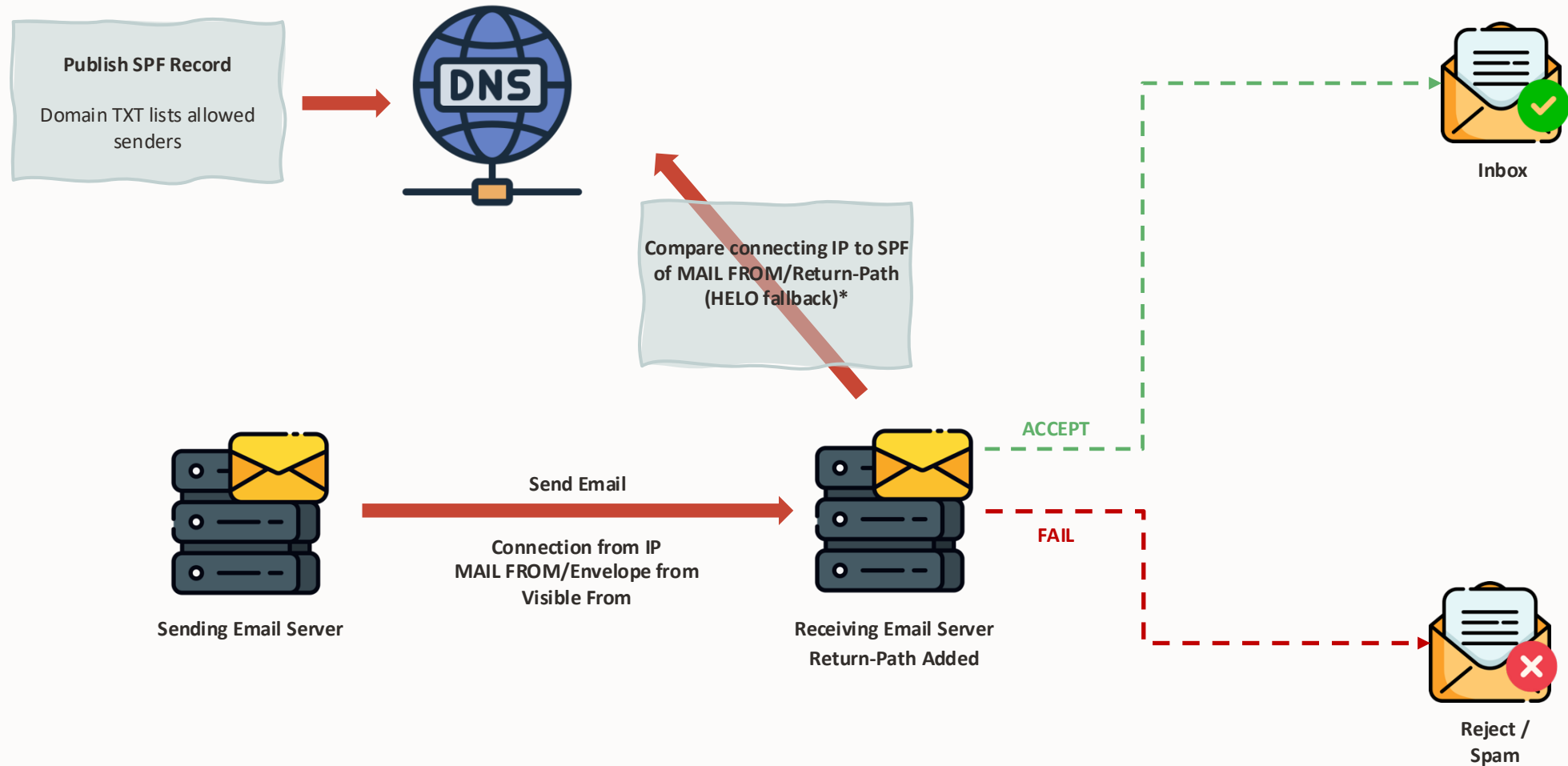
DKIM (DomainKeys Identified Mail)



* DKIM aligns the d= domain

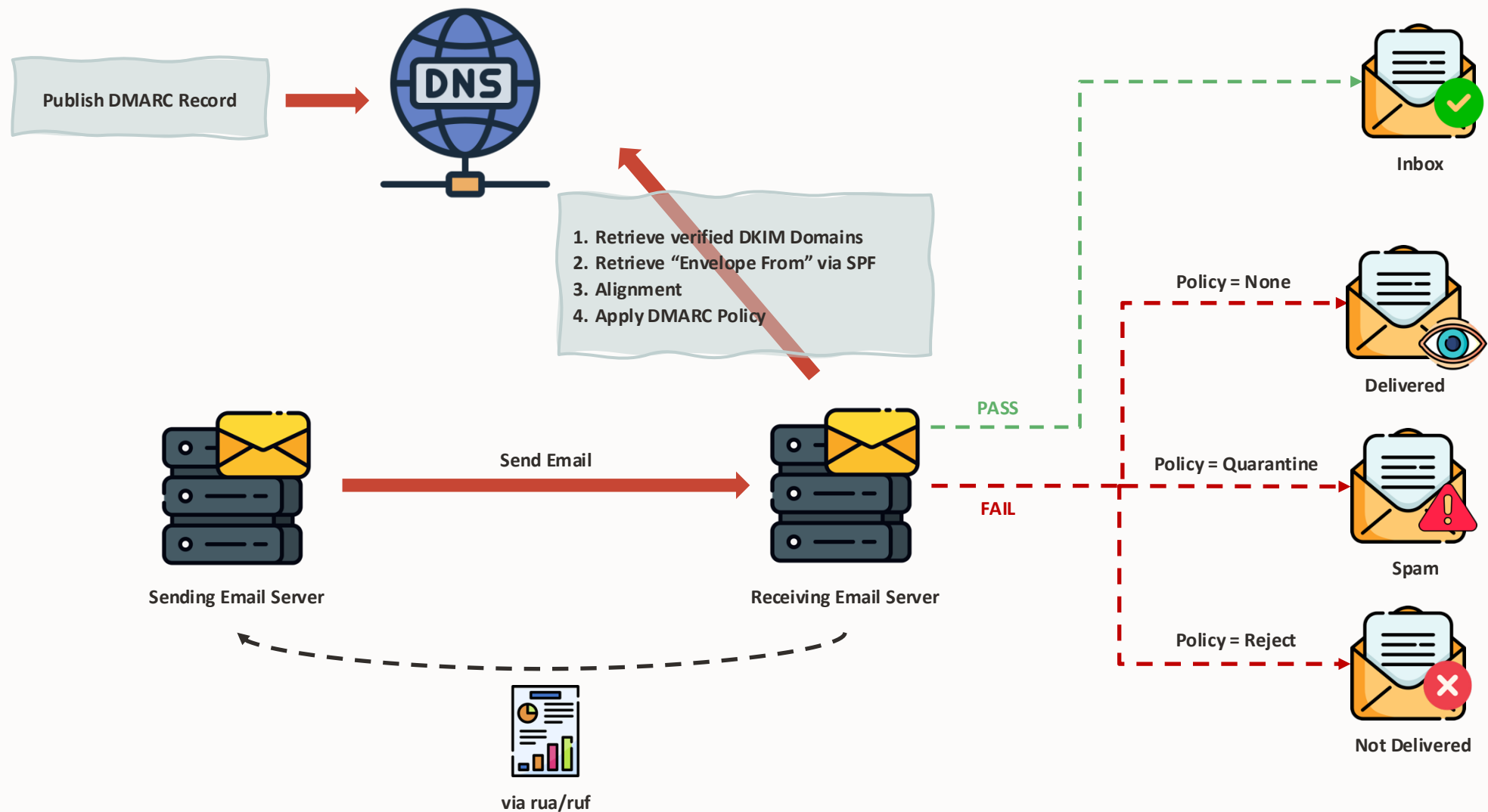


SPF (Sender Policy Framework)

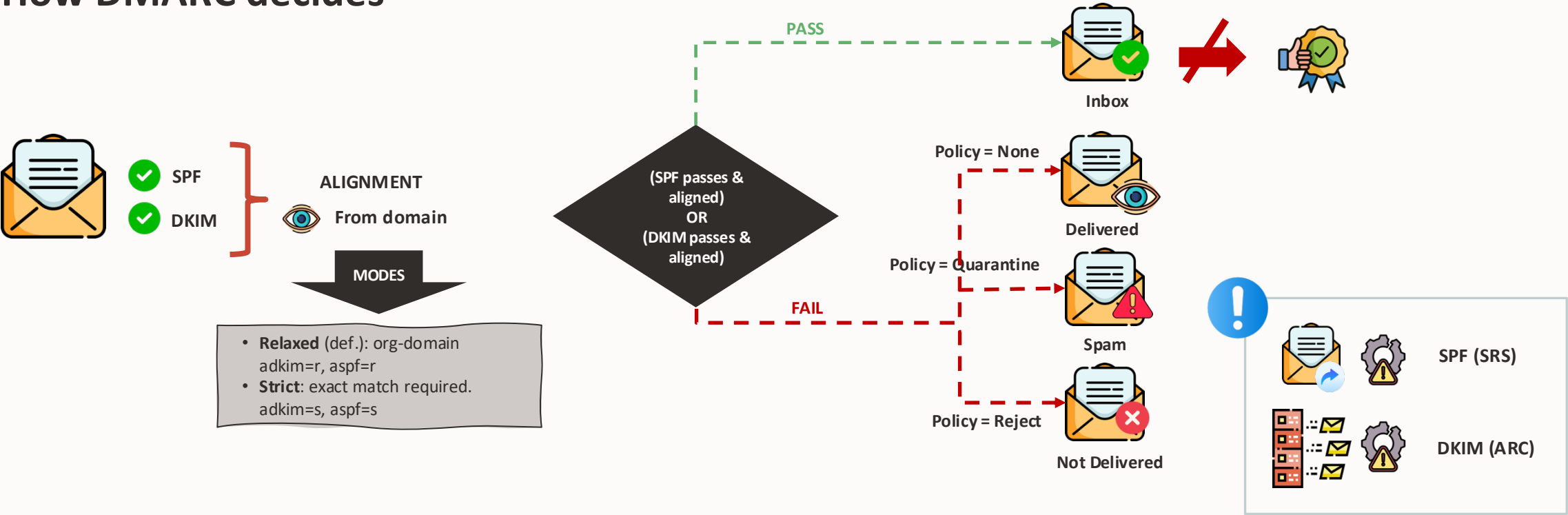


* SPF aligns against RFC5321.MailFrom (Return-Path); receivers may fall back to HELO/EHLO only when MailFrom is null

DMARC (Domain-based Message Authentication, Reporting & Conformance)



How DMARC decides



DMARC Example entry for acme.com domain

- To create a DMARC policy, a DNS TXT entry is created. The domain starts with `_dmarc`, followed by the From domain. The RDATA specifies the policy

Domain	Type	TTL	RDATA
<code>_dmarc.acme.com</code>	TXT	3600	<code>v=DMARC1; p=reject</code>

[DMARC Description](#)



DKIM/SPF/DMARC – OCI Setup

1. .

1 Approved Sender: approve each exact From (resource is regional).

2 Email Domain + DKIM (recommended): add selectors, publish Console CNAMEs, verify Active.

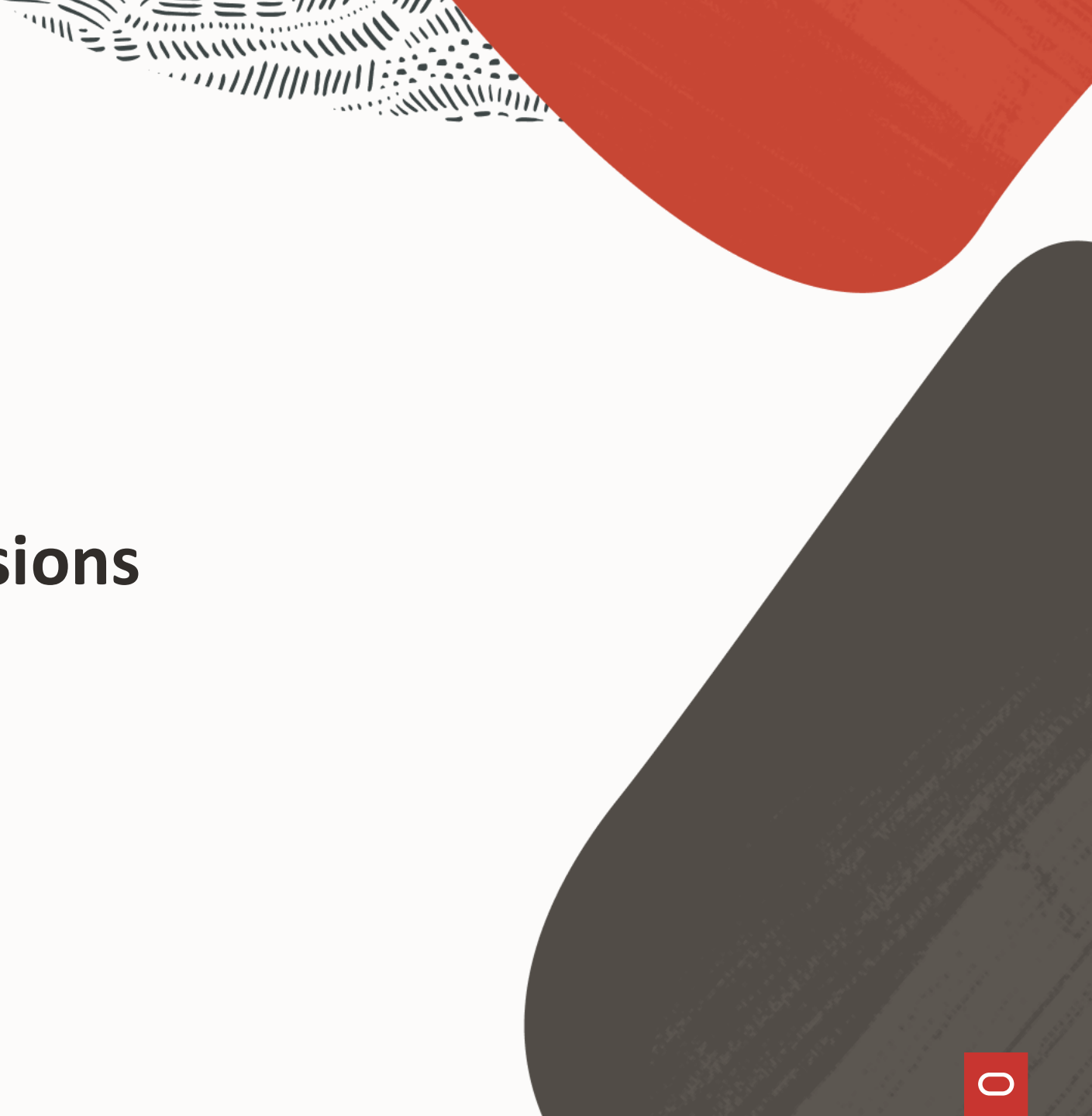
3 SPF (single TXT at apex): include the OCI sender for your realm/region.

- **Americas:** *include:rp.oracleemaildelivery.com*
- **APAC:** *include:ap.rp.oracleemaildelivery.com*
- **EU:** *include:eu.rp.oracleemaildelivery.com*

4 DMARC (on the From domain): a message passes DMARC if DKIM or SPF passes in alignment with the From domain.

- Start with `p=none; rua=mailto:...` to monitor, then move to quarantine/reject once all sources align. (Default alignment is relaxed: `adkim=r, aspf=r`.)

5 (Optional) Custom Return-Path: use your own subdomain for MAIL FROM / Return-Path to strengthen SPF/DMARC alignment and branding



Setting Up Email Submissions

Sending Mail HTTPS and HTTPS raw

HTTPS (Email Data Plane)

- Submit email via REST API or official SDKs (Java, Python, Go, TypeScript/JS, .NET).
- Auth with API key/instance principals.
- Recommended for modern apps.

HTTPS Raw

- Great for complex HTML/inline images when you control headers (headers + MIME) as-is

Prerequisites

- Approved Sender or Email Domain (DKIM) configured
- SPF/DMARC recommended for deliverability.

Transport note

- **Client → OCI:** TLS required.
- **OCI → recipient:** opportunistic TLS when the recipient supports it.

OCI CLI

- **Control plane:** “oci email ...” (approved senders, email domains, DKIM, suppressions).
- **Submission:** “oci email-data-plane ...” (send over HTTPS).
- **Submission:** “oci email-data-plane submit-raw-email ...” (send over HTTPS raw).

Sending Mail SMTP

SMTP submission

- Use regional “host smtp.email.<region>.oci.oraclecloud.com” on port 587 with STARTTLS (TLS 1.2).
- Requires SMTP credentials on an OCI user ((limit 2 per user)
- From must be an Approved Sender (regional). When using Email Domains + DKIM, create the approved sender in that domain so it’s verified.

Prerequisites

- Approved Sender (required).
- Email Domain + DKIM recommended (enables verification)
- SPF/DMARC recommended for deliverability.

Transport note

- **Client → OCI:** TLS required.
- **OCI → recipient:** opportunistic TLS when the recipient supports it.



Managing Email domains, Approved Senders, Supression List, DKIM, Custom Return Path

Managing Email Domains, Approved Senders, DKIM and Suppression List

Email Domains

- Create Email Domain → generate DKIM → publish DNS → verify
- Send from any address at your domain; rotate DKIM keys periodically
- Optionally configure Custom Return Path per region

Approved Senders

- Authorize specific From addresses (regional resource)
- Good for quick starts or tightly scoped apps
- Still publish SPF/DMARC to improve deliverability

DKIM

- Adds a cryptographic signature; proves integrity and ties to signing domain (d=)
- Required for Email Domains; keep From aligned for DMARC
- Rotate keys and re-verify regularly

Suppression List

- Auto-added after hard bounces or complaints; further sends are blocked
- View / add / remove via Console or API
- Do not re-send to suppressed addresses; investigate list quality
- Track EmailsSuppressed along with bounces and complaints

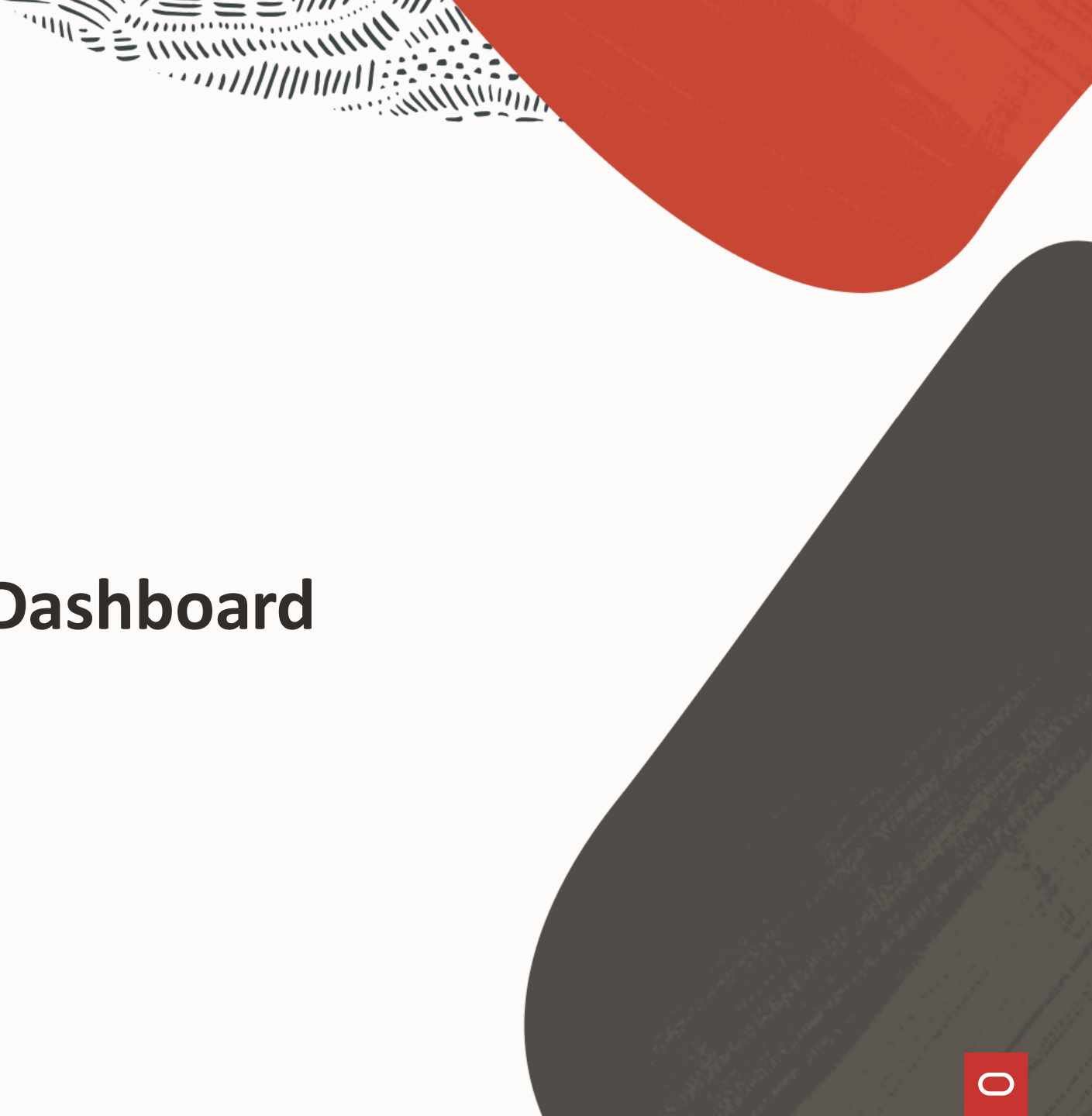
Managing Custom Return Path

What & Why

- Return-Path = bounce address used by receivers/services
- Custom Return Path brands MAIL FROM with your own domain
- Enables SPF-based DMARC alignment (DMARC can pass even if DKIM fails)
- Improves deliverability, reputation, and traceability
- Only one active Custom Return Path per Email Domain (per region)

When to use

- Production environments, high-volume senders, regulated industries
- When enforcing DMARC (p=quarantine or p=reject)
- When you need to build your own sender reputation instead of relying on Oracle's shared domain
 - If not: Higher spam probability as ISPs see this as less trustworthy configuration



Email Deliverability and Reputation Governance Dashboard

Email Deliverability dashboard

IAM Policies

Allow group **MetricReaders** to [**inspect** | **read**] metrics in compartment ABC.

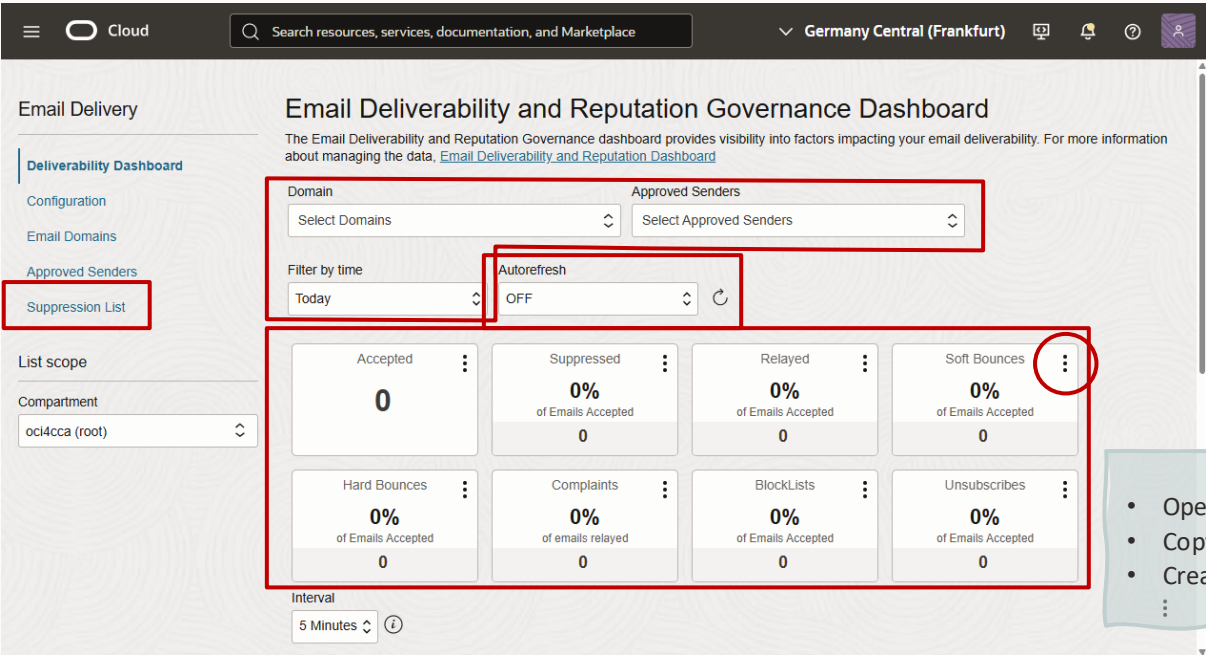
OCI Console

Menu → Developer Services → Application Integration → Email Delivery → **Deliverability Dashboard**

- Tenancy-wide
- Auto-adds after hard bounce, complaint, or one-click unsubscribe; future sends blocked.
- Manage via Console/API (use sparingly; require re-opt-in).

Email Domain Logs

- Enable OutboundAccepted + OutboundRelayed
- Trace via messageId/recipient



Metrics: oci_emaildelivery

- EmailsAccepted
- EmailsSuppressed
- EmailsRelayed
- EmailsSoftBounced
- EmailsHardBounced
- EmailComplaints
- EmailsBlocklist
- EmailsListUnsubscribed

- Open chart/table
- Copy MQL
- Create an alarm



Limits Counting & Cost

Service limits (per region) + Counting rules

Resource ¹	Limit Name	Universal Credits / PAYG	Trial	Always Free ²
Emails sent in last 24 hours	<i>max-emails-day</i>	50,000	200	0
Maximum approved senders	<i>approved-sender-count</i>	10,000	2,000	10
SMTP credentials per user		2		2
Emails sent per minute	<i>sendrate</i>	18,000	10	10
Message size (encoded) ³	<i>max-message-size</i>	2MB	2MB	2MB

[Email Delivery Service Limits Documentation](#)

Counting & Pricing

- An email is defined as either a single recipient (as defined in the To:, CC:, or BCC: fields) or a 2 MB chunk of an email.
- The email volume limit applies to unique recipients among all emails sent.
 - 1 email request with 10 recipients (TO:, CC:, or BCC:) = 10 mails
 - A 10 MB email sent to a single recipient = 10MB/2MB = 5 mails
 - 1 email request with a message size of 10 MB and 10 recipients = (10MB/2MB) x 10 recipients. = 50 mails
 - 1 single mail + 100 recipients = 100 mails + 1 single recipient
- Billed per 1,000 emails (accepted/suppressed), regional scope.

¹ Limits can't be increased without SPF and DKIM set up for each sending domain.

² Per tenancy and cannot be increased

³ Size can be increased till 60 MB.



Compartments, Quotas & Scope

Limits are enforced tenant-wide per region (daily volume + sendrate)

QUOTA



- You can apply quotas to specific limit names
- For Email Delivery, the documented quota is approved-sender-count at regional scope (control how many senders exist)

QUOTA



- They don't subdivide the tenant's daily volume or sendrate per compartment.
- Use IAM + quotas to control who can create/use senders, not how much they can send.

Governance example

- If the region's daily cap is 50,000, usage in one compartment consumes from the same 50,000 regional pool for the tenant.

Compartments, Quotas & Scope

Administrators can set Compartment Quotas to divide tenancy wide limits across compartments

Example:

Tenancy-wide limit for max-emails-day is 50000

A compartment quota of 40000 has been set for both compartment A and B.

When 40000 emails have been sent from compartment A, compartment B can still send 10000 emails.

Policy Example

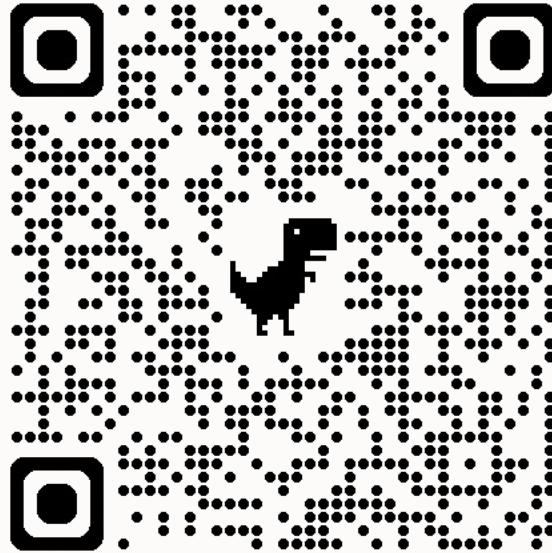
- Set email-delivery quota max-emails-day to 1000 in compartment <compartment_name> where request.region = eu-frankfurt-1

Documentation

- [Overview of Email Delivery](#)
- [Email Delivery Service Capabilities and Limits](#)
- [Getting Started](#)
- [Overview of Compartment Quotas](#)
- [Configuring Oracle and Third-Party Applications](#)



Stay tuned



[Operations Advisory](#)
[GitHub Repository](#)

Thank you

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